Forecasting With Honest Uncertainty

Research summary for the Condor Forecaster: methods, validation, and data

Condor Funds v2 · internal research brief · August 22, 2026 · condensed from three full reports in `docs/research/` (methods ladder, validation, data sources)

Executive summary

We plan a Forecaster: given a portfolio, project wealth 1–5 years forward as a fan chart with 65% and 95% bands, plus a "project from two years ago and compare" backtest view. Three research passes were run — a survey of forecasting methods from naive to state-of-the-art (with measurements on our own cached price data), a survey of how probabilistic forecasts are validated honestly, and an inventory of free data sources that could feed richer models. The findings converge on one theme:

> **The dominant uncertainty at multi-year horizons is not market randomness — it is that we do not know the expected return.** The standard error of an estimated annual return depends only on the calendar span of the data (Merton 1980): with 10 years of history and 18% volatility it is ±5.7 percentage points *per year*. Every sophistication upgrade we evaluated (GARCH, regime switching, machine learning) buys less accuracy at a 2-year horizon than simply admitting this one error bar — which costs nothing to include.

Recommended build, in order: (A) closed-form constant-rate bands ("GBM") with a μ-uncertainty overlay, drawn as two nested band sets so the user sees market randomness and estimation error separately; (B) a stationary block bootstrap for realistic path shapes, guard-railed so it can never present narrower bands than A; (C) a visible "expected return assumption" control anchored by long-run references (ties into the existing Black-Litterman backlog item). **Skip the ML tier entirely** — a 2-year forecast trained on 10 years of data has five independent observations to learn from, and 2026 benchmarks show even foundation models barely beat a random walk on financial series.

1 · Three kinds of not-knowing, one formula

A fan chart answers "what could my wealth be in T years?" We can fail to know this for three distinct reasons, and they are not the same size:

Source	What it is	Grows with horizon like
Path uncertainty	Even with the true return and dispersion known, the realized sequence is random	$SD \propto \sqrt{T}$
Parameter uncertainty	μ and Σ were estimated from a finite window and are wrong by a knowable amount	$SD \propto T$ for μ — **linearly**
Model uncertainty	The "future resembles the past" assumption itself; regimes shift, companies change	Unbounded; not quantifiable from the data

The first is what every naive Monte Carlo tool shows. The second is what nearly every consumer tool omits, and it grows *faster* than the first. The third is why the chart needs a sentence of prose next to it, forever.

Under i.i.d. returns with annual volatility σ, estimation window N years, horizon T years, the variance of cumulative log return is approximately $\sigma^2 T$ (path) + $T^2\sigma^2/N$ (μ-error). Two consequences organize everything:

- **Sampling frequency is irrelevant to μ.** Ten years of daily data and ten years of monthly data give the *same* standard error on the mean: $\sigma/\sqrt{N}$ per year. Finer sampling improves $\hat{\sigma}$ only. (This also resolves our parked sampling-interval question for the forecaster: daily data is right for dispersion; no window length rescues μ.)
- **Band-width inflation from admitting μ-error is $\sqrt{1 + T/N}$:**

Horizon	N = 5 y window	N = 10 y	N = 20 y
1 year	×1.10	×1.05	×1.02
2 years	×1.18	**×1.10**	×1.05
5 years	×1.41	**×1.22**	×1.12
10 years	×1.73	×1.41	×1.22

Measured on SPY from our own price store (2016–2026): $\hat{\mu}$ = 14.3%/yr, $\hat{\sigma}$ = 18.0%/yr, so SE($\hat{\mu}$) = 5.7 points. A 95% confidence interval on the expected annual return runs roughly **3% to 26%**. Concretely, for $10,000 over 2 years: path-only 95% range $8,130–$22,000; with estimation error $7,720–$23,310. The width moves ~12%; more importantly, **the location of the entire fan — the median line — is only known to about ±11 points of annual return**. A tool that prints "median outcome: $13,300" without that caveat is overstating what it knows.

Two further technical notes worth keeping: (i) compounding at the arithmetic mean overstates median wealth by $\exp(\sigma^2 T/2)$ — 3.3% at σ=18%, T=2y; 25% for a 30%-vol single name over 5 years — so the projection must use the log/geometric drift; (ii) μ and σ have different sufficient statistics, so it is legitimate (and almost never offered by consumer tools) to estimate σ from a short recent daily window and μ from the longest defensible span or an anchor.

2 · The method ladder, rung by rung

Rung 0–1 — Constant-rate ("GBM") bands: analytic, then Monte Carlo

Assume i.i.d. returns; wealth is lognormal and the quantiles are a closed-form formula — no simulation needed. Fat tails are a red herring at multi-year horizons for a diversified portfolio: summing ~500 daily returns is close to normal regardless (measured: GBM vs bootstrap 95% bands on SPY differ by under 2% at 2 years). **Verdict: build first.** It is instant, it is explainable, and it is the closed-form pin every simulated method gets verified against — exactly our house testing style.

Rung 2 — Bootstrap of actual history (i.i.d., then block)

Resample our own daily returns instead of assuming a distribution; block/ stationary variants (Politis–Romano) keep runs of days together so volatility clustering and drawdown sequences survive. Two traps, both measured on our data:

- **Automatic block-length selection misfires on returns.** The standard Politis–White rule, applied to raw returns, picks 1–6 day blocks (returns are serially near-uncorrelated) — which silently degenerates to the i.i.d. bootstrap. Applied to *absolute* returns (where the clustering lives) it picks 60–85 days. Practical answer: fix ~21-day blocks, disclose it.
- **Block bootstrap can dishonestly narrow the bands.** It reproduces whatever mean-reversion is in the sample — and at 2-year scale a 10-year window contains *four* independent observations of that. Measured variance ratios at 2 years across our own tickers: 0.05 (SPY) to 2.37 (XOM) — pure noise. On SPY the 63-day block bootstrap produced 28% *narrower* 2-year bands than i.i.d.: an artifact of one recovered-every-dip decade, not a discovery about markets.

Verdict: build second, for path realism ("what a bad two years looks like"), with a guard rail: if its bands come out narrower than rung A's, show the wider ones.

Rung 3 — Parameter-uncertainty overlay (the one that matters)

Draw each simulated path's μ from its sampling distribution $N(\hat{\mu},\ \hat{\sigma}^2/N)$ — or fold it into the closed form, where it stays analytic. Cost: essentially zero. Effect at 2 years: +10–12% band width, and the honest admission that the centre line is an estimate. This is the frequentist twin of the Bayesian posterior predictive under a flat prior, so we get the "right" answer without any MCMC machinery. (Σ-uncertainty, by contrast, is ~1.4% relative with a decade of daily data — negligible for the fan; it matters for the frontier, not the forecast.) **Verdict: ship with rung A from day one.**

Rung 4 — Volatility dynamics: EWMA, GARCH, filtered historical simulation

GARCH-class models forecast *near-term* volatility well, but the effect decays with a half-life of ~2–7 weeks; today's volatility state contributes at most a fixed number of days' worth of extra variance no matter the horizon. Measured with typical equity persistence: starting from *double* the long-run volatility widens the 2-year band by ~14% and the 5-year band by ~6% — the extreme case buys about what the free μ-overlay buys in the normal case. **Verdict: defer.** Worth revisiting only for a 1–12-month view or a "current market conditions" toggle (e.g. reopening the app the week after a crash).

Rung 5 — Regime switching (Markov)

Conceptually attractive (it is our parked "piecewise process" intuition made formal) and the literature (Guidolin–Timmermann) finds real regime structure in returns. But over a 2-year simulation the chain visits its long-run mixture anyway; fits on 10-year windows are unstable; and it adds $K\times(K+1)$ parameters of exactly the estimation risk nobody prices. **Verdict: skip for now** — the block bootstrap delivers most of the tail-shape benefit non-parametrically. Revisit if a "market regime" concept ever becomes a product feature of its own.

Rung 6 — Bayesian: priors and anchors for μ

The genuinely valuable Bayesian move is not machinery but a **prior on μ**: a 10-year bull-market sample mean (SPY: 14.3%/yr) is not a defensible 2-year expectation, and shrinking toward a long-run/equilibrium anchor (7–8%) both centres the fan somewhere honest and narrows the μ error bar by bringing in outside information. PyPortfolioOpt's Black–Litterman implementation (already installed) does this properly and closes an existing backlog item at the same time. Full posterior sampling (PyMC) adds nothing for the conjugate case and would force a Python upgrade. **Verdict: rung C — a visible "expected return assumption" control** (Historical X% / Long-run market Y% / Custom), which is also the most educational widget the feature can have: the user watches the whole fan hinge on that one number.

Rung 7 — The ML tier: quantile boosting, conformal prediction, deep and foundation models

The binding constraint is arithmetic, not architecture: a 2-year-ahead target from 10 years of history has **five non-overlapping training examples** (overlapping windows are 99.6% redundant). Conformal prediction — whose entire selling point is honest coverage — needs many sequential feedback rounds; at 2-year horizons we get one round every two years. The 2026 evidence on time-series foundation models (Chronos-2, TimesFM-2.5, Moirai-2.0) applied to equities finds gains over a random walk "small and sparse," statistically significant in 2 of the tested cases. In the M6 forecasting competition, 8.6% of professional teams beat the naive benchmark with significance; the winner improved on it by 2.2%. **Verdict: skip.** The right place to spend effort is quantifying how badly we know μ, not on a better μ.

Ladder at a glance (10k paths, 2-year horizon, measured runtimes)

Method	Adds	Cost	Runtime	Verdict
Analytic GBM bands	the baseline + test pin	S	instant	**Build (A)**
μ-uncertainty overlay	the dominant error, honestly	S	free	**Build (A)**
i.i.d. / block bootstrap	real tails, drawdown paths	S–M	~0.2 s	**Build (B)**, width-guarded
μ anchor / Black–Litterman	defensible centre line	M	instant	**Build (C)**
GARCH / FHS	near-term vol state	M	~0.4 s	Defer (short-horizon view only)
Regime switching	state-conditional tails	M–L	slow fit	Skip for now
PyMC full Bayes	nothing beyond closed form here	L	minutes	Skip (needs Py ≥3.12 too)
Conformal / boosting / DeepAR / TSFMs	—	M–L	—	Skip: ~5 independent 2-y observations

3 · Validating a fan chart honestly

Our house rule is that every engine function gets a verification test. For a probabilistic forecast, the validation splits into three layers with very different strengths, and knowing which layer can support which claim is most of the honesty.

What a fast, deterministic test suite can pin (CI)

- **Closed form:** simulated GBM quantiles must match the analytic lognormal quantiles within Monte-Carlo tolerance; the μ-overlay band must widen by $\sqrt{(1+T/N)}$.
- **Consistency:** on synthetic i.i.d. data with known moments, bootstrap bands must converge to the analytic ones; GBM and bootstrap must agree with each other on data generated by a GBM.
- **Implementation checks of the scoring/calibration tools themselves** (coverage tests reject at ~nominal rates on synthetic data).

None of this says the forecast matches reality — it says the engine computes what it claims to compute. That distinction should be explicit in the code.

What a periodic offline notebook can measure (real data, quarterly)

- **Proper scoring rules** — metrics that cannot be gamed by making bands dishonestly narrow or wide: **CRPS** as the headline number (generalizes MAE to distributions; closed form for lognormal, so it scores rung A directly), pinball loss per band quantile to localize *which part* of the fan is off, and the interval (Winkler) score as the one-number summary of each band. Python: `scoringrules`.
- **Calibration** — "the 95% band should contain reality 95% of the time" is testable: hit-rate tests (Kupiec), clustering of misses (Christoffersen), and the **PIT histogram** (transform each realized outcome through its forecast CDF; the result should be uniform — U-shaped means bands too narrow, humped means too wide, sloped means the centre is biased). This is how the Bank of England's fan charts are audited.
- **Walk-forward design** — monthly forecast origins, expanding and rolling windows, and the asset universe frozen as of each origin date (using today's-survivors is look-ahead in disguise). Results stratified by sub-period, so a method tuned on 2010–2020 that fails 2022 is visible.

> **The long-horizon wall.** With ~25 years of usable data, a 2-year horizon yields ~12 non-overlapping outcomes; a 5-year horizon yields 5. No test has real power at that sample size, overlapping windows recover apparent-but-not-actual sample size (a "63% hit rate on the 65% band" carries a standard error of roughly ±15 points), and the standard corrections (Newey–West / Hansen–Hodrick) are known to be most biased precisely when horizon is long relative to sample. Formal calibration claims are supportable at horizons up to ~1 year; beyond that, credibility must come from *verified arithmetic plus short-horizon calibration*, descriptive overlays of the few real outcomes we have, and stress replays (2000–02, 2008–09, 2020, 2022). No one can validate a 5-year band statistically — including everyone who sells one.

What the UI may say — and must not say

- Supportable: "this range reflects historical variability in returns, estimated from your lookback window"; "backtested" only for ≤1-year coverage claims.
- Avoid: a bare "95% probability" attached to a multi-year band (implies a validated claim that cannot exist); any suggestion the bands capture regime change or events outside the training history; unhedged confidence in the median path — μ is the least-precisely-estimated input in the whole system.
- If we ever show a "probability of reaching a goal," show it as a *range across return assumptions* — single probability scores swing wildly on half-point changes in assumed return.

4 · Data sources worth wiring in (all free)

Source	What it unlocks	Terms	Effort
1. Fama-French factors (Ken French library, monthly to 1926, via `pandas-datareader`)	Factor-based μ shrinkage (a far better prior than one sample mean) and a factor covariance model — the biggest modeling payoff available	OK (de-facto standard; attribute)	Low
2. FRED regime series (yield-curve slope T10Y3M, credit spread BAA10Y, VIX/VIX3M — same client we already use)	Widen/narrow bands by regime. *Not* point-forecast drivers: Goyal–Welch (2008; 2024 follow-up) shows macro predictors almost never beat the historical mean out of sample — but they are well-supported *risk* signals	OK (attribution, generous limits)	Trivial
3. Shiller CAPE + Damodaran implied ERP (direct spreadsheet downloads; 1871– and 1960s–)	Long-horizon expected-return anchors — exactly what rung C's "expected return assumption" control should offer as its non-historical options	OK (attribute)	Low
4. SEC EDGAR XBRL (audited fundamentals, ~2009–, 10 req/s, no key)	Ground-truth fundamentals; the cleanest license of any source reviewed	OK (public data)	Moderate (tag normalization)

Notable negatives: **no reliable free source of single-name implied volatility exists in 2026** (realized vol from our own store is the fallback, and is fine); Alpha Vantage / Polygon / EODHD free tiers are rate-limited into uselessness (~5–25 requests/day-to-minute); Nasdaq Data Link's useful free sets are gone. yfinance/Tiingo fundamentals remain ToS-gray — acceptable at our scale, not something to build a customer-facing feature on. Factor/macro series don't fit the ticker-keyed `PriceSource` protocol; they get a sibling `FactorSource` protocol with a long cache TTL when we wire them in.

5 · What we will build

1. **Rung A (first ship):** closed-form constant-rate fan with the μ-uncertainty overlay, presented as two nested band sets — inner "market randomness," outer "+ uncertainty in our return estimate" — with the estimate's error bar printed as a sentence ("Estimated return 14.3%/yr; from 10 years of data that is

good to about ±5.7 points"). Instant to compute; verified against the lognormal closed form. Clearly labeled in the UI as the simplest model.

2. **Rung B:** stationary block bootstrap (fixed ~21-day blocks, disclosed), for path realism and drawdown narratives; guard-railed to never show narrower bands than A.
3. **Rung C:** the expected-return-assumption control (Historical / long-run anchor / custom), Black–Litterman underneath, CAPE/ERP as anchor sources.
4. **Backtest view** at every rung: refit on data ending two years ago, draw the fan, overlay what actually happened, report the percentile it landed in — with the sample-of-one caveat attached, and the offline coverage study (not the UI) carrying any statistical claims.

UI honesty checklist, distilled: name what the median assumes; show both band sets; print the μ error bar as a number; dollars, not log returns; disclose the knobs (method, window, block length, paths); one line on the blind spot ("these bands assume the future resembles the past decade in kind; they cannot price a change in what your holdings fundamentally are"); never a bare multi-year probability.

Full reports with references, measured tables, and package assessments: `docs/research/forecast-methods-ladder.md` (methods, with measurements on our price store), `docs/research/forecast-validation.md` (scoring rules, calibration tests, backtest design), `docs/research/forecast-data-sources.md` (source-by-source terms and integration notes). Key external references: Merton (1980); Pástor & Stambaugh (2012); Jacquier, Kane & Marcus (2005); Politis & Romano (1994) and Patton–Politis–White (2009); Goyal & Welch (2008) and Goyal, Welch & Zafirov (2024); Gneiting & Raftery (2007); the M6 competition papers (2023–24). Agent-produced research; citations spot-check before external use.